

Data to Design

Linear Model Family Report

S1 Linear Family

July 2026

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1 Results and Interpretation

1.1 IID Baseline Results

This subsection presents the baseline performance of the Linear Model Family on the UCI Concrete dataset and the UHPC dataset. The tested models were Ordinary Least Squares (OLS), Elastic Net, Bayesian Ridge, and Polynomial Ridge.

1.1.1 UCI Dataset

The UCI experiments used an 80/20 train-test split. Hyperparameter tuning was performed with 4-fold cross-validation, following the four-part validation idea used in Yeh’s original concrete strength study.

Original Dataset					Domain-Ratio Dataset				
Model	MAE	RMSE	R	R^2	Model	MAE	RMSE	R	R^2
OLS	8.72	10.94	0.790	0.617	OLS	8.45	10.69	0.803	0.634
Elastic Net	8.75	10.95	0.790	0.616	Elastic Net	8.58	10.76	0.803	0.629
Bayesian Ridge	8.75	10.95	0.790	0.616	Bayesian Ridge	8.49	10.71	0.803	0.633
Polynomial Ridge	4.69	6.57	0.929	0.862	Polynomial Ridge	4.72	6.71	0.926	0.856

Log-Transformed Dataset					Interaction Dataset				
Model	MAE	RMSE	R	R^2	Model	MAE	RMSE	R	R^2
OLS	4.88	6.48	0.931	0.865	OLS	4.49	6.19	0.938	0.877
Elastic Net	4.87	6.48	0.932	0.866	Elastic Net	4.47	6.18	0.939	0.878
Bayesian Ridge	4.86	6.48	0.932	0.866	Bayesian Ridge	4.48	6.18	0.939	0.878
Polynomial Ridge	4.03	5.80	0.946	0.892	Polynomial Ridge	3.78	5.66	0.949	0.897

Table 1: Linear-family baseline and feature-representation results on the UCI Concrete dataset.

Summary

Dataset	Best Model	RMSE	R^2
Original Dataset	Polynomial Ridge	6.57	0.862
Domain-Ratio Dataset	Polynomial Ridge	6.71	0.856
Log-Transformed Dataset	Polynomial Ridge	5.80	0.892
Interaction Dataset	Polynomial Ridge	5.66	0.897

Interpretation The plain linear models were too simple for the UCI dataset, staying near RMSE 10.9 and $R^2 \approx 0.62$ on the original variables. Polynomial Ridge performed much better because it can represent curved effects such as age gain and nonlinear mix-ratio behavior. The interaction dataset was the strongest setup, and across three random seeds the Polynomial Ridge model achieved $\text{RMSE } 5.17 \pm 0.49$ and $R^2 = 0.901 \pm 0.007$.

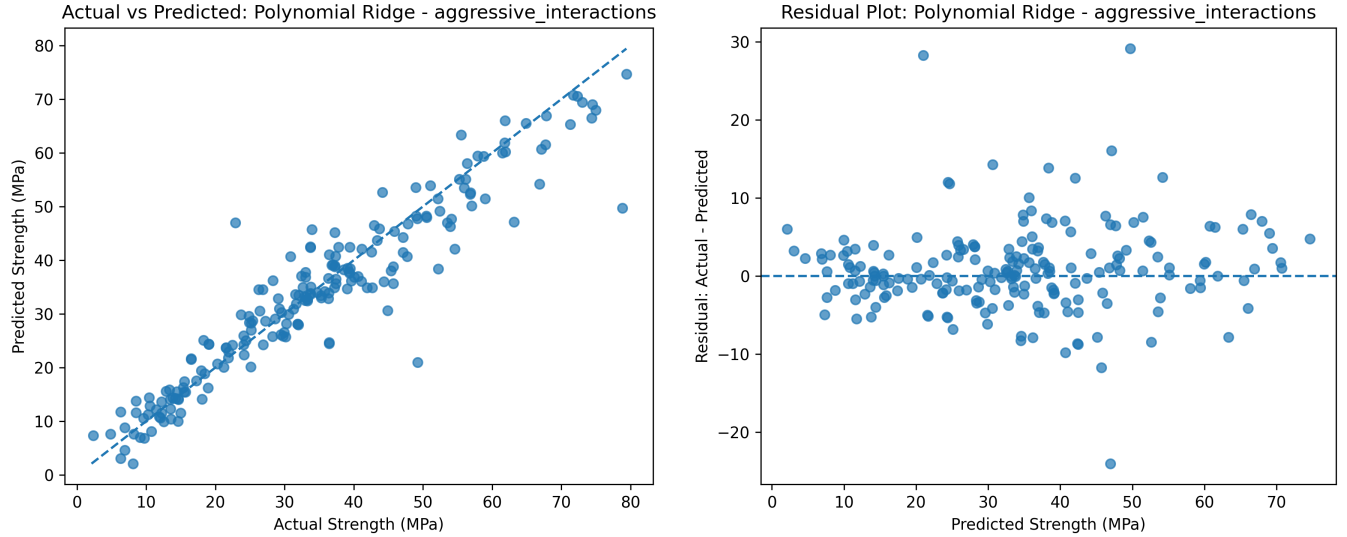


Figure 1: UCI Polynomial Ridge predictions and residuals on the held-out test set.

1.1.2 UHPC Dataset

Dataset Variant The final UHPC modeling setup used the shared semantic recoding strategy with a 50% missingness threshold. After keeping rows with valid 28-day compressive strength, 2073 rows remained.

Dataset	Rows	Raw Predictors	Numeric	Categorical
Source data with publication metadata	2073	34	24	9
Modeling data without publication	2073	33	24	9
Encoded model matrix	2073	60	–	–

Final Model Performance (Random Split) The row-mixed split used 1449 training rows, 311 validation rows, and 313 test rows. Identical feature vectors were kept in the same split to reduce direct duplicate leakage.

Model	Hyperparameters
OLS	{}
Elastic Net	alpha=0.01, l1_ratio=0.3
Bayesian Ridge	alpha_1=1e-05, lambda_1=1e-07
Polynomial Ridge	degree=2, alpha=100.0

Model	CV RMSE	Validation			Test		
		RMSE	MAE	R^2	RMSE	MAE	R^2
OLS	–	21.96	16.97	0.662	21.09	16.56	0.685
Elastic Net	23.97	21.75	16.98	0.669	20.91	16.54	0.690
Bayesian Ridge	24.03	21.74	16.99	0.669	20.93	16.56	0.689
Polynomial Ridge	19.83	343.34	30.65	-81.54	15.51	11.14	0.829

Interpretation Elastic Net and Bayesian Ridge were the most reliable UHPC models. Polynomial Ridge had the best test score, but its validation error was extremely unstable, so it was not selected as the main UHPC model. Bayesian Ridge had the lowest stable validation RMSE, while Elastic Net was retained for the publication-held-out and attribution analyses because it was sparse, stable, and easy to interpret.

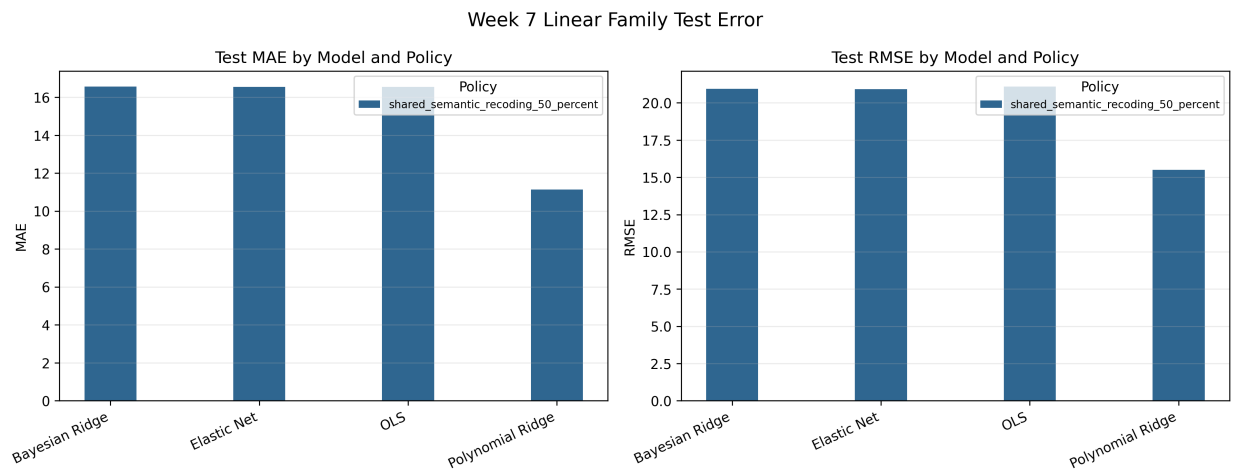


Figure 2: Row-mixed UHPC performance for the Linear Model Family.

1.2 Targeted Experiments

This subsection checks how sensitive the UHPC linear models are to fibre information, outliers, engineered ratios, and multicollinearity.

Removing All Fibre Columns All fibre amount, type, length, and diameter features were removed. This reduced the predictive power of the stable linear models.

Model	Base RMSE	No-Fibre RMSE	Base R^2	No-Fibre R^2
OLS	21.09	24.08	0.685	0.589
Elastic Net	20.91	23.94	0.690	0.594
Bayesian Ridge	20.93	23.97	0.689	0.593
Polynomial Ridge	15.51	20.36	0.829	0.706

Outlier Sensitivity The lower and upper 2.5% of training targets were removed, reducing the training set from 1449 to 1375 rows. This did not improve generalization.

Model	Base RMSE	No-Outlier RMSE	Base R^2	No-Outlier R^2
OLS	21.09	22.41	0.685	0.644
Elastic Net	20.91	22.35	0.690	0.646
Bayesian Ridge	20.93	22.64	0.689	0.636
Polynomial Ridge	15.51	16.45	0.829	0.808

Engineered Ratio Features Adding domain ratios did not improve the stable models. Elastic Net test RMSE changed from 20.91 to 21.39, and Bayesian Ridge changed from 20.93 to 21.41. This suggests that the existing composition variables already carried most of the useful linear information.

VIF Diagnostic The numeric predictors did not show severe multicollinearity. The maximum VIF was 3.81, the median VIF was 1.60, and no feature had VIF above 5.

1.3 Publication-Level Generalisation

Random row splits can be optimistic because rows from the same paper may share materials, equipment, and reporting style. Therefore, the UHPC model was also tested by holding out complete publications.

Publication-Held-Out Split

Split	Rows	Publications	Target Mean	Target Std.
Train	1451	115	145.91	34.59
Validation	311	25	152.21	34.95
Test	311	25	168.15	40.08

Row-Mixed vs. Publication-Held-Out Test

Evaluation	Rows	Model	MAE	RMSE	R^2
Row-mixed test	313	Elastic Net	16.73	21.21	0.681
Publication-held-out test	311	Elastic Net	25.79	30.22	0.430

LOPO Summary

Aggregation	Publications	Rows	MAE	RMSE	R^2
Micro average over rows	6	452	24.82	29.74	0.160
Equal publication average	6	452	24.75	29.61	-1.045

Interpretation The publication-held-out result is much weaker than the row-mixed result. RMSE increased by about 9 MPa and R^2 dropped from 0.681 to 0.430. This shows that the linear model transfers less well to unseen publications than the row-mixed score first suggests.

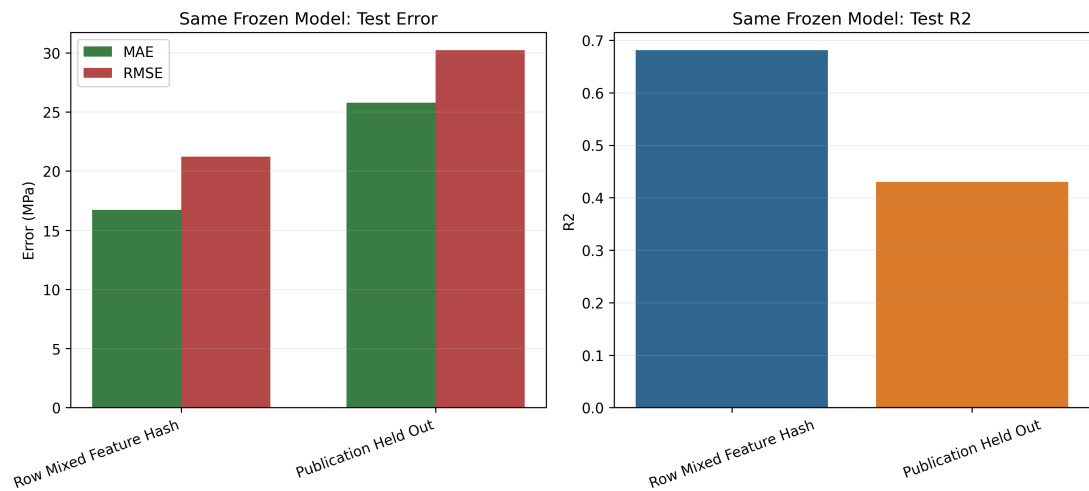


Figure 3: Generalisation gap between row-mixed and publication-held-out testing.

1.4 Uncertainty and Calibration

Prediction intervals were evaluated at the 90% nominal coverage level on the publication-held-out test set.

Method	Model	Coverage	Width	RMSE	R^2
Split conformal	Elastic Net	0.894	94.46	28.49	0.493
Native interval	Bayesian Ridge	0.830	85.11	27.72	0.520
Conformalized interval	Bayesian Ridge	0.936	108.02	27.72	0.520
Residual bootstrap	Elastic Net	0.743	66.94	28.49	0.493
Bootstrap conformalized	Elastic Net	0.916	95.15	28.49	0.493

LOPO Calibration Summary

Method	Model	Coverage	Width	RMSE	R^2
Split conformal	Elastic Net	0.938	101.90	29.46	0.176
Native interval	Bayesian Ridge	0.754	77.80	30.31	0.128
Conformalized interval	Bayesian Ridge	0.931	101.29	30.31	0.128
Residual bootstrap	Elastic Net	0.777	70.63	29.46	0.176
Bootstrap conformalized	Elastic Net	0.925	99.57	29.46	0.176

Interpretation The conformal methods were closest to the desired 90% coverage. However, the intervals were wide, usually around 95–102 MPa. This means that the model can express uncertainty, but predictions for unseen publications should still be treated carefully.

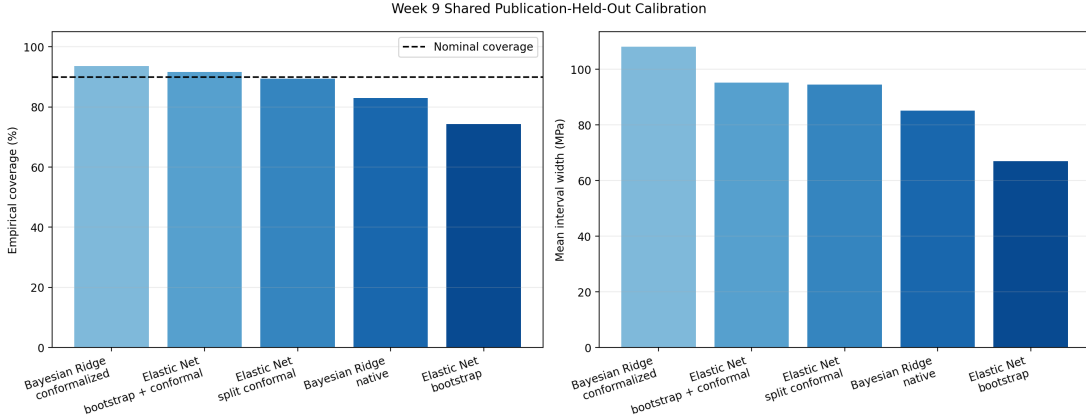


Figure 4: Coverage and interval width for the publication-held-out uncertainty methods.

1.5 Combined Error, Uncertainty and Explanation Analysis

This subsection applies to the UHPC dataset only. SHAP and permutation importance were computed for the final Elastic Net model on the publication-held-out test set. Grouped SHAP is reported as the sum of individual feature mean-absolute SHAP values inside each physical group.

Readiness Check The attribution audit passed: the test rows, targets, and lineage rows were aligned (311/311/311), publication metadata was not present in the predictors, SHAP additivity error was 2.84×10^{-14} , and permutation importance reused the fitted model without refitting.

Table 2: Elastic Net — Grouped Feature Attribution vs. Permutation Importance

Feature Group	SHAP Mean	Permutation Mean RMSE	Permutation Std
Binders and SCMs	21.4954	0.8616	0.4303
Curing	13.8516	8.5186	0.8489
Fillers and Aggregates	11.0776	2.2498	0.2913
Fibres	10.0854	3.8381	0.5073
Chemical Admixtures	3.8011	0.2495	0.1789
Water	2.7843	0.5347	0.1202

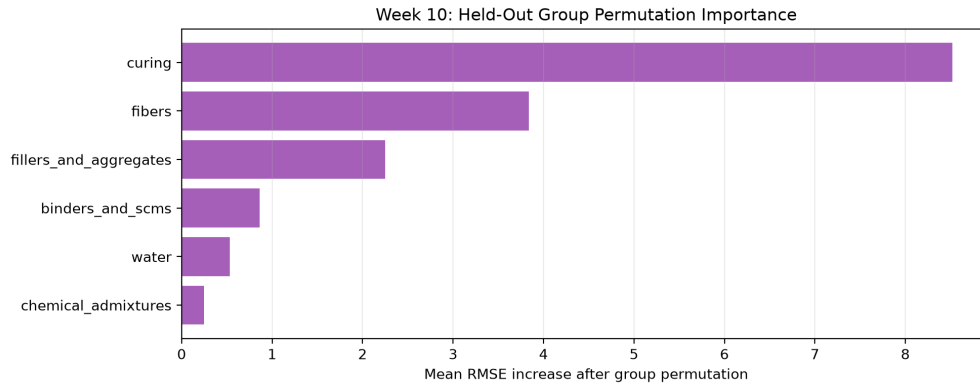
Elastic Net: Grouped Feature Attribution vs. Permutation Importance

Table 3: Elastic Net — Top 10 Individual Raw Features by SHAP Mean

Feature	SHAP Mean	Permutation Mean RMSE	Permutation Std
curing_temp	12.2728	8.5350	0.8266
sand_type	7.6199	1.3389	0.3843
fiber1_amount	6.5705	2.3305	0.4192
fly_ash_type	5.1856	0.5348	0.2892
cement_type	3.4299	0.4082	0.1611
silica_fume	3.3745	1.0121	0.1847
fly_ash	3.1480	0.5071	0.1311
fiber1_type	2.9135	0.5736	0.1390
water	2.7843	0.5399	0.1311
sp_type	2.1885	0.8631	0.1467

Elastic Net: Top 10 Individual Raw Features by SHAP Mean

Interpretation The SHAP and permutation views answer slightly different questions. SHAP assigns large total contribution to binders and SCMs because this group contains many cementitious variables. Permutation importance shows that curing is the most damaging group to shuffle, and `curing_temp` is the strongest individual feature by both SHAP and permutation. Therefore, curing is not a mistake in the implementation; it is genuinely important for the linear model, especially when testing on unseen publications.

**Figure 5:** Grouped permutation importance for the Elastic Net model on held-out publications.

2 Conclusion

Polynomial Ridge worked very well on the cleaner UCI benchmark, especially after interaction features were added. On UHPC, the stable linear models were less accurate but easier to interpret. The most important finding is the generalisation gap: row-mixed testing looked acceptable, but publication-held-out and LOPO testing showed that unseen publications are much harder. The final explanations are still physically sensible, with curing, fibres, binders, SCMs, fillers, and aggregates all contributing to model behavior.

References

- [1] I.-C. Yeh, Concrete Compressive Strength Dataset, UCI Machine Learning Repository, 2007.
- [2] I.-C. Yeh, “Modeling of strength of high-performance concrete using artificial neural networks,” *Cement and Concrete Research*, vol. 28, no. 12, pp. 1797–1808, 1998.